

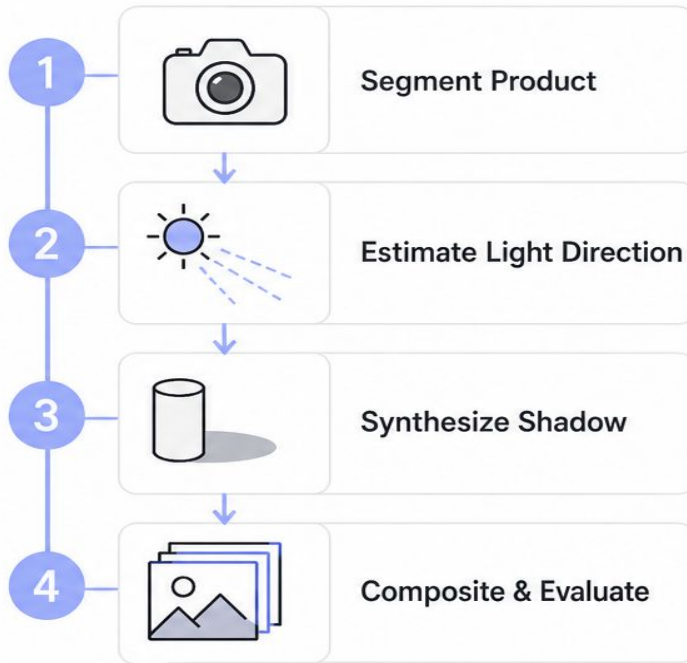
Illumination-Aware Product Photography

Preserving Foreground Lighting • Adapting the Background

Casual Photo (Messy Background)



Illumination-Aware Result (Clean Studio)



Himanshu Jhavar
hj2713 • uni: hj2713

The Problem with AI Product Photography Today

Background generation ignores the product's real-world lighting — and the result looks physically wrong.

❌ What brands do today

1

Casual photo
(Real lighting)



2

AI background
(Wrong shadow)



3

Problem
(Physically wrong)



Shadow points
wrong way

Product lit from right ·
Background shadow on
bottom-right —
physically impossible



✅ What we do

1

Casual photo
(Real lighting)



2

Estimate light
(From product shading)



We read the shading on the
product and estimate the
light direction θ .

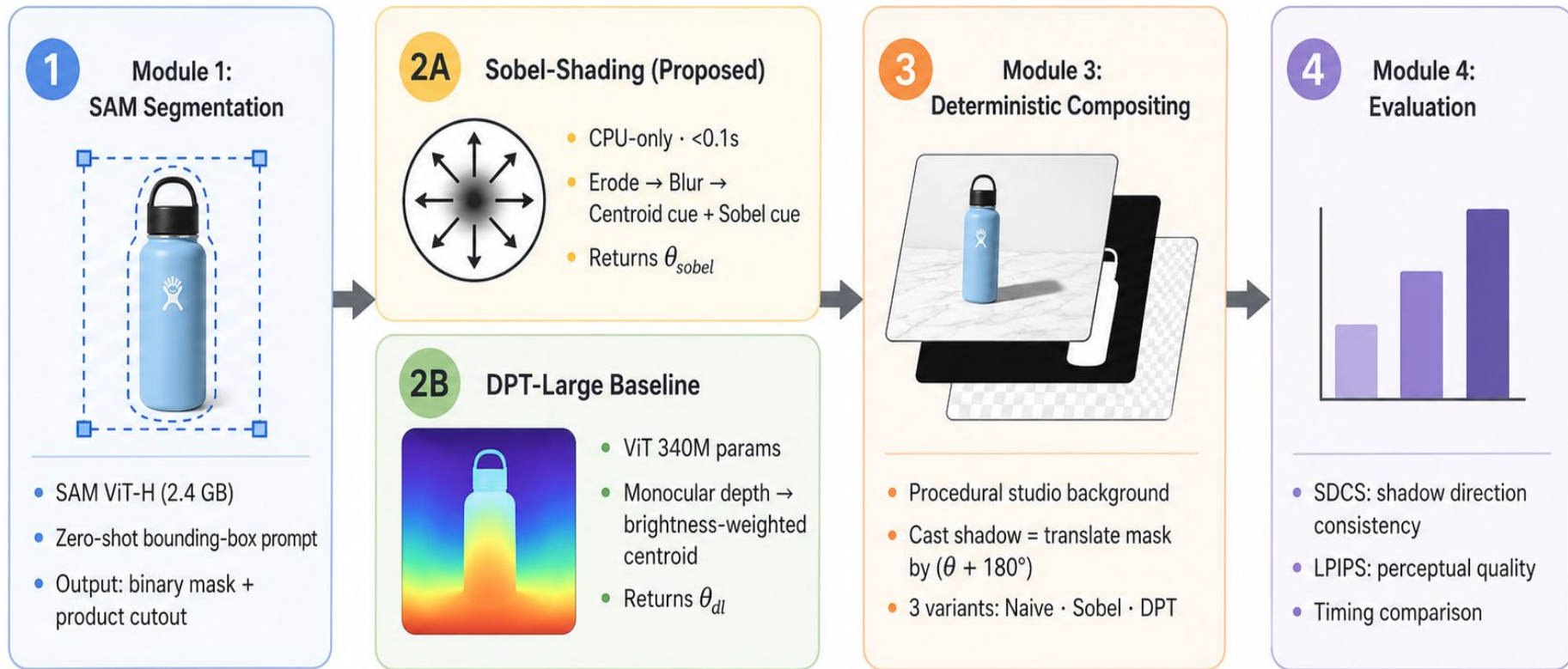
3

Our result
(Correct shadow)



We estimate the light direction from the product itself and enforce physical consistency in the generated scene.

Pipeline Architecture



Conclusions & Takeaways

What We Built



Fast, training-free light estimation — Sobel shading on eroded, blurred product interior. No GPU, no labels, no training data required.

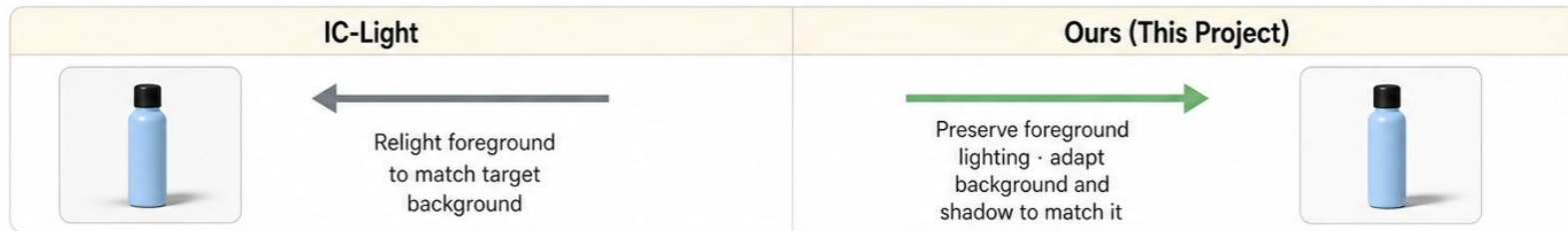


Physically grounded compositing — shadow cast in opposite direction of estimated light, directly from mask geometry. Product pixels preserved exactly.



Architecture pivot — replaced unstable Stable Diffusion inpainting with deterministic CV compositing. Makes the illumination contribution explicit and controllable.

vs. IC-Light: The Inverse Problem



We study the reverse direction of illumination harmonization.

Honest Limitations



Transparent & reflective products degrade estimation quality



SDCS is a heuristic shadow metric, sensitive to background texture and sparse shadows



Single dominant light source assumed — multi-light scenes not modeled



SAM mask quality limits shadow accuracy on low-contrast product edges